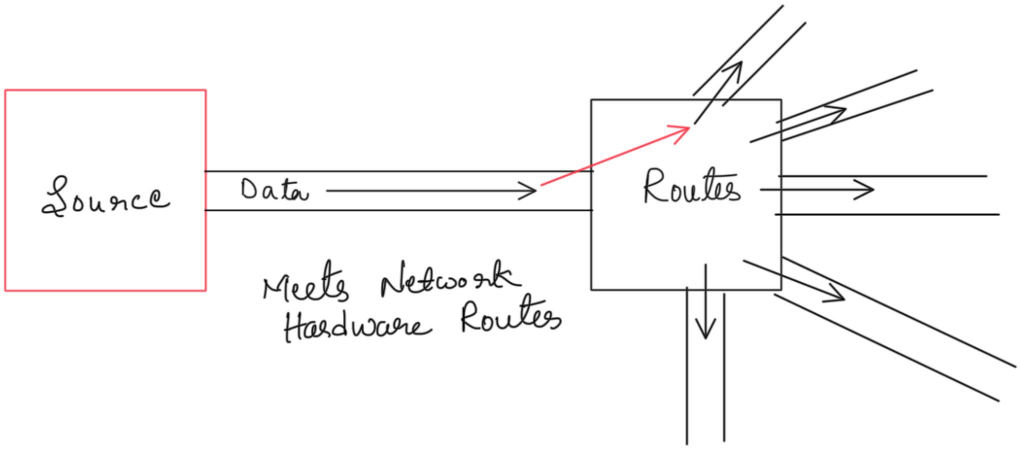
Computer Networks

DAY 87
05/10/2025



Router decides the received packate's next travelling path

The Data packate travel from Source computer to a nearest Router along the destination path.

once the data packate reach the router then the router will decide in which path the next packate will be sent

proxy

What is proxy?

Proxy Server is an intermediary between your device (client) and the Internet (Server)

When we use a proxy

- you dont connect directly to a website.
- You connect to the proxy Server, and the proxy connects to the website on your Behalf

your computer → Proxy Server → Internet (website)

Internet (website) → Proxy Server → your computer

⚙️ 2. Why Is It Used?

Purpose	Description
Privacy	Hides your IP address from the websites you visit.
Security	Filters malicious content, blocks access to dangerous sites.
Content Filtering	Organizations (like schools/offices) use proxies to control which websites can be accessed.
Caching	Proxies can store copies of web pages → speeds up repeated access.
Geo-Unblocking	Lets you appear as if you're browsing from another country.
Logging & Monitoring	Used in enterprises to monitor internet usage.

🧠 3. How It Works (Step by Step)

Let's say you open `www.example.com` through a proxy.

1. You send your request to the proxy (not directly to example.com).
2. The proxy checks rules (like filters, authentication, etc.).
3. It forwards your request to the website.
4. The website responds → proxy receives it → sends it back to you.

So your actual IP stays hidden — the website only "sees" the proxy's IP

🛡️ 4. Types of Proxies

Type	Description
Forward Proxy	Used by clients to access the internet indirectly (common for privacy or filtering).
Reverse Proxy	Sits in front of servers — used by web servers (like Nginx) to manage traffic, load balancing, and caching.
Transparent Proxy	Doesn't modify requests/responses, often used by ISPs or schools.
Anonymous Proxy	Hides your IP address but identifies itself as a proxy.
High Anonymity Proxy	Hides both your identity and the fact that you're using a proxy.

⚡ 5. Real-Life Analogy

Imagine you want to talk to someone but don't want them to know who you are.

So, you send your message through a **middleman (proxy)**:

- You → Proxy → Target person.

  The target only knows the proxy, not you.

That's exactly what happens online with proxy servers.

Without proxy:

```
You → [www.google.com]
Your IP: 192.168.1.5
```

Google logs: `Request from 192.168.1.5`

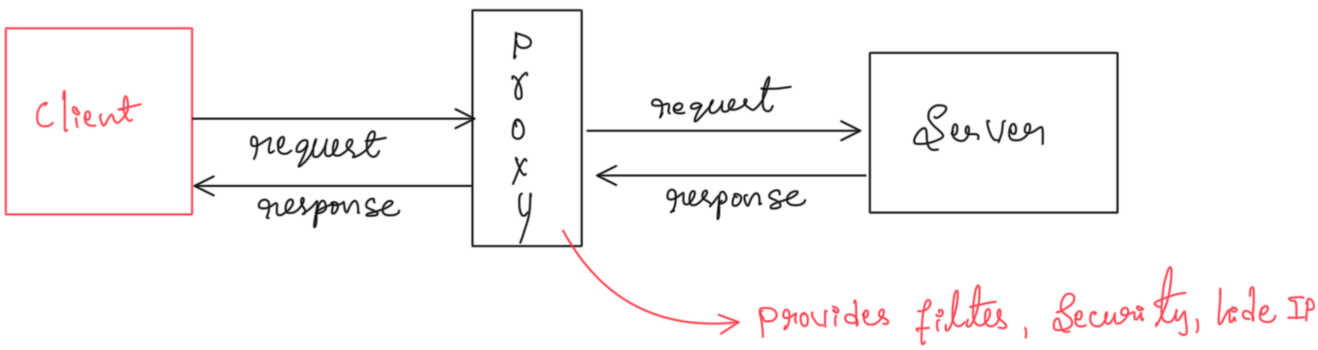
With proxy:

```
You → Proxy Server (45.23.11.7) → [www.google.com]
```

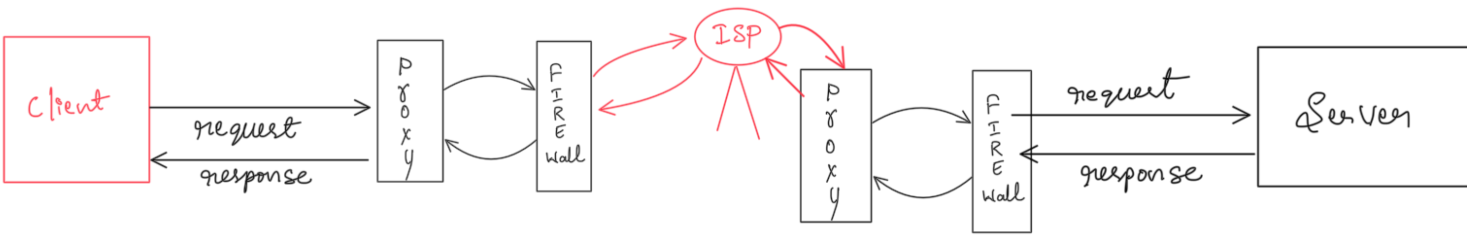
Google logs: `Request from 45.23.11.7`

Proxy knows your identity, but Google does not.

In Companies
Proxies are used for Security, Caching, and internet Control



Fire Wall



1. What is fire wall ?

A firewall is a Security System that monitors and controls incoming and outgoing network traffic based on pre defined Security rules

It acts as a barrier between your trusted network (like your PC or office LAN) and the untrusted network (Internet)

2. Why its called "Fire Wall" ?

The Term Comes from construction
In buildings, a firewall is a physical wall that stops fire from spreading

Similarly in computers, a network firewall stops "bad traffic" (like, viruses, hackers, or unauthorized access) from spreading into your System.

⚙️ 3. What Does It Do?

A firewall:

- 👮 **Monitors traffic** entering or leaving your computer/network.
- 🚫 **Blocks unauthorized access** or malicious traffic.
- ✅ **Allows legitimate communication** (like your browser connecting to a website).
- 📋 **Applies rules** like:
  - "Allow HTTP (port 80) and HTTPS (port 443) traffic."
  - "Block all incoming traffic except from trusted IPs."

Every packate which has Sent Have Sequence

| 3 | 2 | 1 |

packet which is lost that will be tracked and Sent Back

🔒 7. Real-World Analogy

🔄 4. How It Works (Step-by-Step)

When data (packets) enter or leave your system:

1. The firewall **inspects** the packet header (IP address, port number, protocol, etc.).
2. It checks this against a **set of rules** (firewall policy).
3. Based on that:
   - **Allow** the packet (if it matches a safe rule)
   - **Block or drop** the packet (if it's suspicious or unauthorized)

Imagine a **security guard** at a building entrance:

- Checks **ID** (IP address)
- Ensures **authorized persons** enter
- **Blocks intruders** or suspicious people

That's what a firewall does for your network.

🧩 5. Types of Firewalls

Type	Description	Example
Packet-Filtering Firewall	Examines packets based on IP, port, protocol. Simple but fast.	Routers, iptables
Stateful Firewall	Tracks ongoing connections; only allows packets part of valid sessions.	pfSense, Cisco ASA
Proxy Firewall (Application-Level)	Acts as an intermediary between user and Internet. Can inspect app data.	Squid Proxy Firewall
Next-Gen Firewall (NGFW)	Deep inspection + intrusion prevention + malware filtering.	Palo Alto, Fortinet
Cloud Firewall	Protects cloud resources (like AWS, Azure).	AWS WAF, Cloudflare

Each packet Have time to Live (TTL)

Every routes that forwards the packate decreases TTL By 1

When TTL is 0 the packet will be dropped and Sends "Time Exceeded" Error to the Sender (Source)

⚔️ 8. Firewall vs Proxy

Feature	Firewall	Proxy
Main Purpose	Security – block/allow traffic	Privacy, caching, filtering
Works On	Network layer	Application layer
Controls	IP, Port, Protocol	Content (URLs, web requests)
Protects	Network/system	User identity & traffic
Example	Cisco Firewall, pfSense	Squid Proxy, Nginx Reverse Proxy

🧠 9. Why We Need Firewalls

- Prevent unauthorized access.
- Protect against network attacks (e.g., port scanning, brute force).
- Restrict malware communication.
- Enforce company or personal security policies.

🕸️ 4️⃣ What About "Packets Roaming in the Network"?

Sometimes, you might wonder:

💭 "If packets are lost, where do they go? Do they keep roaming forever?"

👉 No — they **don't roam forever**.

Each packet has a **TTL (Time-To-Live)** value in its header (set by sender, like 64 or 128 hops).

🧩 7️⃣ Real-World Analogy

Every router that forwards the packet **decreases TTL by 1**.

When TTL = 0 →

❌ The router **drops** the packet and sends an **ICMP "Time Exceeded"** message back to sender.

So, packets can never loop infinitely.

They're safely **discarded** if they wander too long.

Imagine sending **10 parcels** via courier:

- **TCP**:

  You track each parcel, get delivery confirmation, and resend if lost

  📨 Reliable but slower.

- **UDP**:

  You just drop all parcels at once — no tracking.

  🚀 Faster but risky — if lost, it's gone.

(a) TCP – Transmission Control Protocol

TCP is **reliable** — it has **built-in recovery** mechanisms.

Process:

1. **Sender gives sequence numbers** to packets.
2. **Receiver sends ACKs (Acknowledgments)** for each packet received.
3. If the **sender doesn't get ACK** within a timeout →
   🔁 **It retransmits the lost packet.**
4. **Reordering** is done if packets arrive out of order.

✅ TCP ensures:

No loss, no duplication, and in-order delivery.

Example:

- Browsing websites (HTTP)
- Email (SMTP)
- File transfer (FTP)

🧠 6️⃣ Summary Table

Feature	TCP	UDP
Reliability	Reliable (retransmits lost packets)	Unreliable (no retransmission
Acknowledgment	Yes	No
Speed	Slower	Faster
Use Case	Web, Email, Files	Games, Calls, Streaming
Packet Recovery	Built-in (ACK + retransmission)	Not supported

(b) UDP – User Datagram Protocol

UDP is **unreliable** and **connectionless** — it just sends packets and doesn't care what happens to them.

🚫 No ACK

🚫 No retransmission

🚫 No ordering

If a packet is lost → it's **gone forever**.

Example:

- Video streaming
- Online gaming
- VoIP (calls)

Because speed is more important than perfection.

lost packate Recovery depends on which protocol Being used.